A New Frontier for Single Image Reflection Removal in the Wild

Computer Vision Project Presentation

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Part 1: The Reflection Removal Problem - an Overview

The Problem Statement

Given a single image $\mathbf{I}$ captured through a reflective surface (like glass), estimate:

- Transmission layer $\mathbf{T}$: the actual scene behind the glass
- Reflection layer $\mathbf{R}$: the unwanted reflections on the camera's side

Why is it ill-posed?

- The simplest model: $\mathbf{I} = \mathbf{T} + \mathbf{R}$
- One equation, but two unknowns

Taxonomy of Reflection Removal Methods

To overcome the ill-posed nature of $\mathbf{I} = \mathbf{T} + \mathbf{R}$, different **input modalities** and **physical hardware cues** are used to reduce ambiguity and constrain the solution space.

- Multi-image and motion-based methods
- Polarization-based methods
- Flash-based and interactive methods
- Single image methods

Multi-Image and Motion-Based Methods

- Use a series of photos taken from slightly different positions
- The background (transmission) and the reflections resting on the glass plane lie at distinctly different depth planes
- When a camera moves, they appear to move at different speeds, allowing the network to *peel* them apart
- **Pros:** Highly effective at separation and preserving background details
- **Cons:** Mandates a moving camera or moving subjects, thus unsuitable for instant, static point-and-shoot photography

Polarization-Based Methods

- Exploits the optical physics of light, meaning we consider the wave nature of light
- Reflected light tends to be highly polarized, while transmitted light from the background scene is largely unpolarized
- Images are captured through a rotating polarizing filter at multiple different angles.
- Intensity variations of the reflection across distinct angles are analyzed, allowing the network to mathematically isolate and subtract the reflection component
- **Pros:** Achieves excellent physical separation and robustness
- **Cons:** Requires specialized hardware (physical polarizers or polarized cameras), limiting consumer accessibility.

Flash-Based and Interactive Methods

- **Flash-Based:**
  - Takes two successive rapid photos — one with a flash and one without
  - The flash heavily illuminates the glass surface and reflections but rarely has the power to reach the distant background
  - The model learns the background scene through comparison of the two images
  - **Pros:** Introduces a controlled physical change to the scene that mathematical-only methods lack
  - **Cons:** Flash creates harsh, unnatural lighting artifacts and only works when the glass is relatively close to the camera

- **Interactive (Human-in-the-loop):**
  - The user explicitly marks the reflection or the background using strokes, bounding boxes, or text prompts
  - This acts as a semantic guidance, completely removing the "blind" aspect of the source separation
  - **Pros:** Highly accurate and customizable
  - **Cons:** Require manual, time-consuming user effort per image

Single Image Reflection Removal (SIRR)

- The most practical but most computationally challenging paradigm.
- Highly variable lighting conditions (bright daylight, neon signs at night)
- Diverse physical glass properties (varying thickness, curvature, color tint)
- Regions of extreme reflection intensity (saturated reflection) may completely occlude the background transmission

Different Hypotheses

Linear hypothesis: Posits that a captured image $\mathbf{I}$ is perceived as the superimposition of a transmission layer $\mathbf{T}$ and a reflection layer $\mathbf{R}$

- The simplest model: $\mathbf{I} = \mathbf{T} + \mathbf{R}$
- Blending scalars: $\mathbf{I} = \alpha\mathbf{T} + \beta\mathbf{R}$
- $\mathbf{I} = \alpha\mathbf{T} + (\mathbf{1} - \alpha)\mathbf{R}$

Non-linear hypothesis: Leverages non-linear models learned through deep learning models

- $\mathbf{I} = \mathbf{W} \odot \mathbf{T} + \mathbf{R}$ with a mask $\mathbf{W}$
- $\mathbf{I} = \mathbf{W} \odot \mathbf{T} + (\mathbf{1} - \mathbf{W}) \odot \mathbf{R}$
- $\mathbf{I} = \mathbf{g}(\mathbf{T}) + \mathbf{f}(\mathbf{R})$ for some degradation functions on $\mathbf{T}$ and $\mathbf{R}$

Part 2: The "In-the-Wild" Challenge & Dataset Generation Methodologies

Synthetic Blending

The Methodology:

- Randomly sample two distinct images from public datasets like Flickr
- Apply artificial Gaussian blurring or brightness adjustments to one image (simulating the out-of-focus reflection layer)
- Blend them mathematically: $\mathbf{I} = \alpha\mathbf{T} + \beta\mathbf{R}$.

The Flaw:

- Lacks genuine physical realism: Real reflections are not simply a linear combination
- Completely ignores complex light attenuation, depth-of-field mismatches, spatially varying illumination, and refraction
- Models rarely generalize well to wild scenarios

Manual Glass Removal

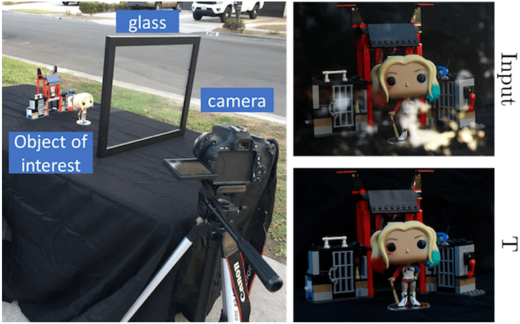
The Methodology (e.g., SIR2, Nature Datasets):

- Set up a camera securely on a tripod facing a scene through a portable glass pane
- Capture the mixed image (**I**) with the glass in place
- Remove the glass manually from the setup and capture the "ground truth" transmission image (**T**)

The Flaw:

- Spatial misalignment due to refraction between **I** and **T**
- Destabilizes pixel-to-pixel reconstruction loss functions during deep learning training, leading to blurred network outputs.

Manual Glass Removal Method



The Paradigm Shift: Stop moving the glass

The Mechanism:

- Use a **non-transmissive, non-reflective black velvet cloth** to physically block one specific light source without touching or shifting the glass
- For example, a black velvet cloth placed carefully behind the glass acts as a light sink, absorbing all transmitted light and perfectly isolating the reflection layer
- Or a black velvet cloth placed carefully on the camera side blocking the local object light isolates the transmission layer

The Black Cloth Method

Zhang et al. [34] | Our M | Our M

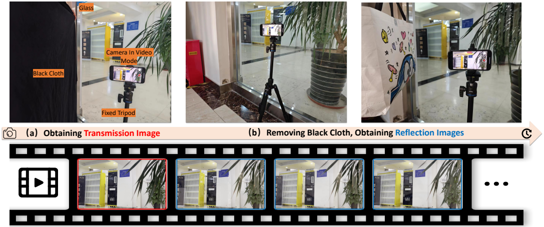
Wei et al. [28] | Our R | Our R

The RRW Dataset Collection Pipeline

Dataset Name: Reflection Removal in the Wild (RRW)

- **Video-Based Capture System:** The entire scene is captured in a continuous high-resolution video mode.
- **The Steps:**
  1. Capture the scene normally with the glass undisturbed **I**
  2. Introduce the black velvet cloth to block local reflection lights
  3. Extract the perfectly isolated transmission frames from the continuous video feed
- **The Advantage:**
  - Logistically easy
  - Perfectly aligned image pairs
  - Completely eliminates the refraction and shifting errors like before

RRW Dataset collection Pipeline



The RRW Dataset: Scale and Impact

- **Scale:** $14,952$ high-resolution, perfectly aligned real-world image pairs
- **Comparison:** Roughly **45 times larger** than previous real-world datasets (e.g., SIR2 454 images, or the Nature dataset 220 images).
- **Impact:** Provides modern, data-hungry deep learning architectures with enough diverse, accurately aligned data to learn

Pipeline Comparison

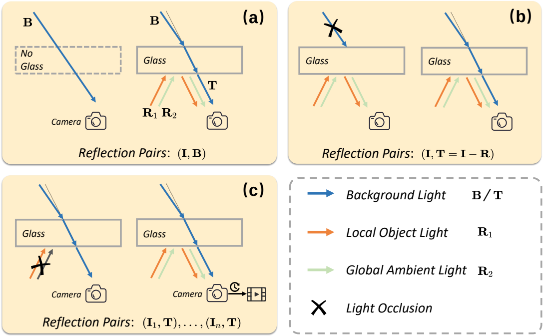


Figure: Existing pipelines for collecting real reflection pairs (I: reflection image). (a) the pipeline may lead to the misalignment between B and T due to the glass refraction. (b) the pipeline is only suitable for collection in the RAW data format, and the obtained T may contain reflection remnant artifacts. (c) Ours, which avoids interference from glass refraction or artifacts and does not require additional data format specifications. Moreover, the video-based capture system would help to reduce the difficulty and effort involved in large-scale data

Datasets Comparison

Comparison of existing important synthetic/real reflection removal datasets

Dataset	Year	Usage	Pair Number	Average Resolution	Real/Syn
CEILNet	2017	Train/Test	7643/850	224 x 224	Syn
RID	2017	Train	3250	224 x 320	Syn
SIR^2	2017	Test	454	540 x 400	Real
Real	2018	Train/Test	89/20	1152 x 930	Real
Nature	2020	Train/Test	200/20	598 x 398	Real
CDR	2021	Train	1063	1542 x 1638	Real
SIR^{2+}	2022	Train	1700	540×400	Real
CID	2023	Train/Test	4800/1200	224×288	Syn
RRW	2024	Train	14952	2580×1460	Real

Part 3: Network Architecture & Loss Formulations

The Cascaded Pipeline

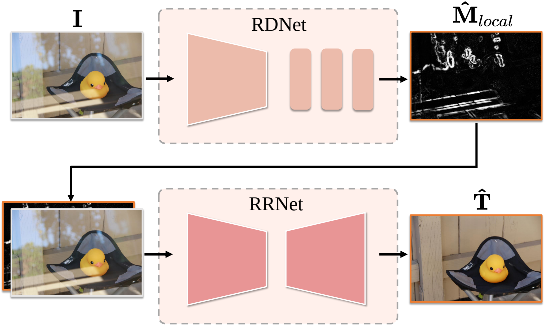
Inspired by cascaded refinement techniques (e.g., Li et al., *SIRR through Cascaded Refinement*), the pipeline is decomposed into two guided sub-tasks:

1. **Reflection Detection Network (RDNet):** Learns to explicitly map exactly where reflections are occurring in the spatial domain.
2. **Reflection Removal Network (RRNet):** Uses the spatial location map generated by RDNet as a structural guide to execute the final restoration of the clean transmission layer.

Data Flow:

$$I \rightarrow \text{RDNet} \rightarrow \hat{M}_{local} \rightarrow \text{RRNet} \rightarrow \hat{T}$$

Data Flow



RDNet Architecture: EfficientNet Backbone

Reference: *Tan et al., "EfficientNet: Rethinking Model Scaling"*

- To accurately predict the spatial reflection mask ($\hat{M}_{local}$), RDNet requires a robust, multi-scale feature extractor capable of distinguishing large reflection blobs from fine background details.
- **Why EfficientNet?** It utilizes compound scaling to optimally balance network depth, width, and resolution.
- This allows it to extract highly accurate semantic and structural features without incurring an excessive computational bottleneck—which is crucial for the first stage of a two-stage pipeline.

Loss Formulation 1: Total Variation (TV) Loss

Reference: *Rudin et al., "Nonlinear total variation based noise removal algorithms"*

- **Goal:** Supervise the predicted reflection mask $\hat{M}_{local}$.
- **The Problem:** Direct pixel-to-pixel loss functions can result in a predicted map that is noisy, speckled, and filled with isolated false positives.
- **The Solution:** We apply a TV Loss alongside L1 Loss:

$$\mathcal{L}_{DNet} = ||M_{local} - \hat{M}_{local}||_1 + \gamma_1 * TVLoss(\hat{M}_{local})$$

- **Effect:** By explicitly penalizing large, sudden numerical differences between adjacent pixels, TV loss enforces **spatial smoothness** and continuity, ensuring the structural guidance passed to the RRNet is coherent.

RRNet Architecture: NAFNet

Reference: *Chen et al., "Simple Baselines for Image Restoration" (NAFNet)*

- Once $\hat{M}_{local}$ is cleanly predicted, it is concatenated with the original mixed image I and fed into RRNet.
- **Nonlinear Activation Free Network (NAFNet):**
  - Replaces traditional, heavy activation functions (like ReLU or GELU) with simple multiplication operations (*SimpleGate*).
  - **Why?** Avoiding aggressive non-linear activations helps preserve the complex linear combinations of light necessary for superior deblurring and reflection removal. It drastically reduces computational complexity while exceeding SOTA performance.

Loss Formulation 2: Perceptual Loss (VGG-19)

References: *Zhang et al., "SIRR with Perceptual Losses" & Simonyan et al., "VGG Networks"*

- Exclusively comparing the raw pixels of predicted $\hat{T}$ and true T (via MSE) often leads to a "regression to the mean" blurring effect.
- **The Solution:** Both images are passed through a fixed, pre-trained **VGG-19** deep feature extractor.
- **Mechanism:** The loss minimizes the distance between the *deep semantic feature maps* of the images rather than just the color values.
- **Result:** Actively forces the network to preserve complex structural details, sharp edges, and textures, ensuring visual realism.

Part 4: Testing Metrics

Quantitative Evaluation: Standard Metrics

To rigorously prove performance, researchers rely on objective quality metrics (e.g., *Hu & Guo, Component Synergy*):

- **Peak Signal-to-Noise Ratio (PSNR):**
  - Measures the absolute, pixel-by-pixel mathematical difference.
  - $PSNR = 10 \cdot \log_{10}(\frac{MAX_I^2}{MSE})$
  - Extremely sensitive to slight spatial misalignments (hence the need for the perfectly aligned RRW dataset).
- **Structural Similarity Index Measure (SSIM):**
  - Perception-based model evaluating perceived degradation in structural information (luminance, contrast, structure).
  - High SSIM confirms background edges and textures were recovered without being destroyed by the algorithmic removal process.

Alternative / Supporting Metrics

While PSNR and SSIM are standard, modern SIRR research utilizes additional metrics to measure deep perceptual quality:

- **LPIPS (Learned Perceptual Image Patch Similarity):** Uses deep network activations to measure perceptual distance. It correlates much more heavily with human visual judgment than PSNR. (Lower is better).
- **NCC (Normalized Cross-Correlation):** Used in cascaded network literature to measure the linear correlation between the estimated layer and ground truth. It is highly robust to global brightness shifts caused by the glass attenuation.

Part 5: Codebase Implementation & Results

Notebook Flow: Consolidating Theory and Practice

Our codebase is structured into interactive Jupyter Notebooks to provide transparent, reproducible testing of the pipeline:

- `00_Visualizing-the-pipeline`: Demonstrates the data stream.
- `01_Benchmark-Evaluation`: Runs the inference loop and evaluates performance across datasets, calculating PSNR/SSIM.
- `02_MaxRF`: Computes gradient maps using the Sobel operator ($G_I > G_T$) to physically visualize how local reflection masks are mathematically generated.
- `03_Analysis-Freq-Domain`: Transforms spatial images into the frequency domain using a 2D Discrete Fourier Transform (DFT). It validates that backgrounds maintain high-frequency structures, while reflections dominate low-to-mid frequency spectrums.

Success Cases: Where the Model Shines

When the model successfully separates the layers, it exhibits high structural fidelity and accurately preserves the transmission layer.

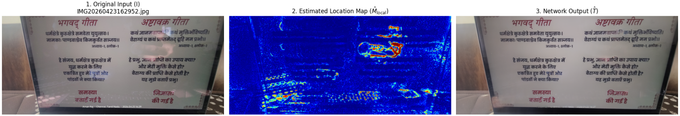


Figure: Consolidated Plot: Successful Reflection Removal (High SSIM)

- Excellent handling of low-frequency, out-of-focus reflection blobs.
- High perceptual quality; underlying text and fine textures are perfectly recovered.

Failed Cases: Where the Model Struggles

Based on our frequency domain analysis, the network sometimes fails to separate specific frequency bands, resulting in distinct visual artifacts.

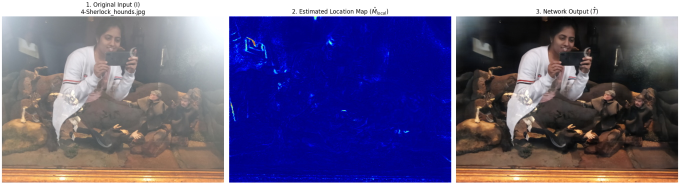


Figure: Consolidated Plot: Network Failure (Ghosting & Smear Artifacts)

- **Ghosting Around Edges:** The network "over-smooths" trying to scrub the reflection, damaging high-frequency background edges.
- **Color Smear/Saturation:** Fails to separate strong, saturated light sources, leaving residual frequency errors behind.

Thank You! Q & A